

Dexamethasone-based modulation of Systemic Cytokines: A Data Analysis

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Date: October 14, 2026

Introduction

The COVID-19 pandemic, produced by SARS-CoV-2, was responsible for an estimated 7,116,673 reported deaths in the world, with 14.83 million additional unreported deaths estimated by the WHO[1, 2]. Highly-severe classifications of the virus are often associated the hyper-inflammatory response known as Cytokine storm[3].

Cytokine storms, also known as hypocytokinemia, describes the life-threatening stage of a disease in which the body of a patient enters immune-system dysregulation as a consequence of viral spread throughout the body. In these scenario's, circulating cytokines induce systemic tissue damage, leading to life-threatening manifestations of hypoxia, hemorrhage, organ-failure, neurologic damage, and more[4]. Hallmark cytokines related to the conditions include but are not limited to IL-6, IL-10, TNF-α, C-reactive protein and VEGF[4].

COVID-19's correlation with inflammatory immune system dysregulation has led to the anti-inflammatory drug Dexamethasone becoming the standard of care in critically-ill patients. The methods by which Dexamethasone decreases symptoms and mortality is not yet fully understood. The aim of this data-analysis is therefore to quantify the effects of Dexamethasone of systemic cytokine concentrations, in the hopes of unveiling its pharmacological actions in preventing cytokine storms.

Data sample

The dataset [5] utilized for this research sets to look at the plasma cytokine levels of critically ill patients with COVID-19. It allows for the comparison of different cytokines and their levels in patients treated with- or without Dexamethasone.

The TSV file, obtained from open data publishing platform 'Dryad'[6], consists of 19 columns, 38 rows. Each row represents a patient (eg. MVIR1-HS107), whilst each column represents a different quantified cytokine (eg. IL-6 or IL-10) in picograms/microliter - pg/mL. The cytokine values were collected on the day of patient's enrollment in the study.

The research which conceptualized the dataset was performed by Lucile Neyton, Ravi Patel, Aartik Sarma and Gabriela Fragiadakis, all of which are associated with the university of California, San Francisco[5].

Data Analysis Methods

0.1 Data Analysis

Once loaded, the TSV dataset did not specify which patients received which treatment-type. The first step therefore required the addition of a treatment column to the dataset that specified which Patient IDs received Dexamethasone and which did not.

The normality of treated and untreated cytokines were compared utilizing skewness functions and the Shapiro-Wilk test[7].This determined which statistical tests can be applied to the data[7].

The dataset was split per treatment group, after which the Mann-Whitney U test could be applied using the **wilcox.test()** function. With 18 Cytokines, 38 patients, and 2 treatment conditions, the multiple-testing-problem can lead to false positive p-values < 0.05. To correct for this, P values were adjusted using the Benjamini-Hochberg procedure[8].

0.2 Log2 Fold change Graph

The log2 fold change per cytokine was calculated in R using **log2()**, with the standard error propagated from the group means. The log2 fol change results, standard errors, marker functional groups and significance of each cytokine, were exported as a *.csv* file. This was subesequently plotted in Python using the **matplotlib** library to have an overview of modulated cytokine behavior based on treatment. Cytokine bars were grouped by marker function (Pro-inflammatory, Anti-inflammatory, Endothelial Injury, & Tissue damage) as to make visual interpretation easier. The significantly affected cytokines from the Mann-Whitney U test were given asterisks (*).

0.2.1 Significant Cytokine Bar-plots

The means, standard errors, p-values and FDR values of the significant cytokines were exported from R and plotted in Python in the same way. Cytokines that proved significant from the data analysis were plotted in individual bar-plots to deduce quantified p-values and concentration changes. Mean, standard deviation and standard error were calculated for each significant cytokine per group. These values can be used to create the bar plots and standard error bars. P-values from the earlier Mann-Whitney U test as well as the FDR corrected P-values were added to each plot.

Results and discussion

Normality testing using the skewness calculations and the Shapiro-Wilk test showed that cytokine distributions had an average skewness of $\tilde{2}.6$, and that 17 of 18 Cytokines consisted Non-Normal data distributions in at least 1 group. Due to the skewed distributions, cytokine means would be inflated, and thus could not be used in significance testing, thereby excluding the usage of paired T-testing.

The Mann-Whitney U test (t-test alternative for non-normal data) indicated that 3 of 18 cytokines changed significantly (p-value < 0.05) between treatment groups, namely IL-6, IL-10, and IFN-γ. Il-6 decreased on average 5.5 fold, IL-10 increased 2,6 fold, and IFN-γ decreased on average 2.0 fold.

A significance threshold of $\alpha = 0.05$ was used, meaning only cytokines with p-values below this threshold were considered statistically significant ($p < 0.05$). By testing 18 cytokines, we could expect $18 \times 0.05 = 0.9$ cytokines to be falsely positive by chance. By adjusting raw p-values using Benjamini-Hochberg False Discovery Rate (FDR) correction, only cytokines with an FDR-adjusted p-value below 0.05 were considered statistically significant, thus avoiding false positives. All 3 previously significant cytokines retained an adjusted p-value below 0.05, with that of IL-6 being 0.00005, IL-10 being 0.014, and IFN-γ being 0.026 respectively.

0.3 Log2 Fold Change Graph

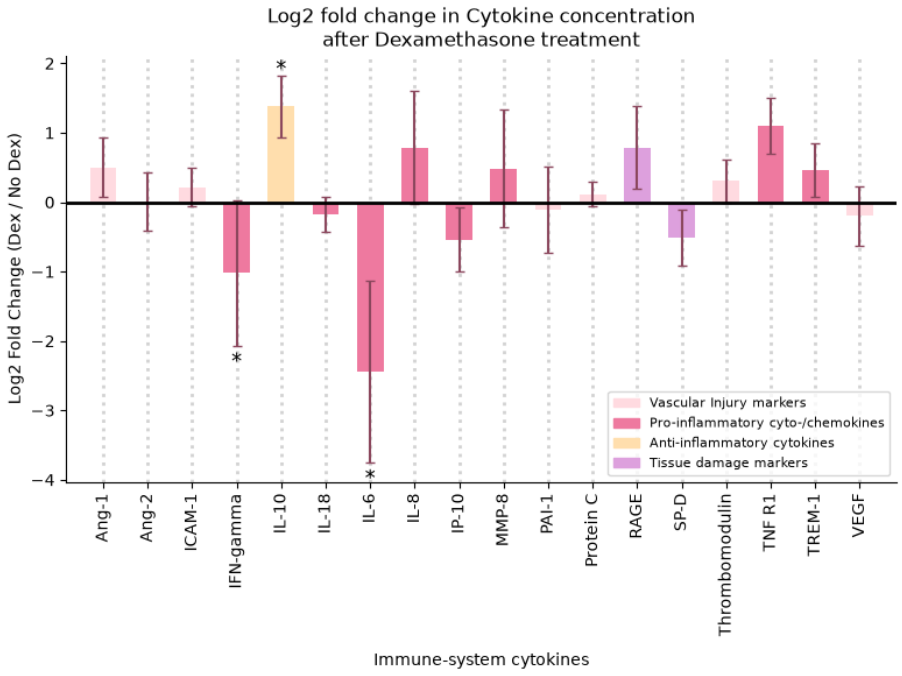


Figure 1: Log2 fold change in cytokine concentrations following Dexamethasone treatment compared to untreated controls. Error bars represent standard error of mean. Bar color was grouped by cytokine functionality. Asterisks were placed by Cytokines with statistical significance (p¡0.005)

0.4 Significant Cytokine Bar-plots

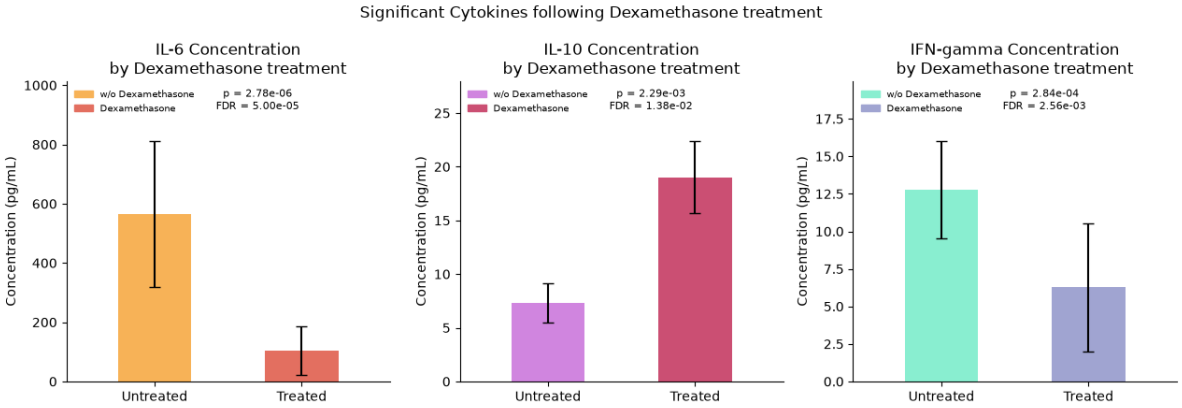


Figure 2: Bar-plots of significant cytokines (IL-6, IL-10 & IFN-gamma) between treated and untreated groups. Each bar-plot contains the respective p values and Benjamini-Hochberg adjusted p values in top-middle of the plot.

The statistical analysis showed that Dexamethasone modulated immune-cell-secreted cytokines (IL-6, IL-10, IFN-γ) as opposed to inhibiting cytokines released by tissue-damage (injury markers such as Ang-1, RAGE, etc). Furthermore, of the three cytokines the affect on pro-inflammatory cytokines (IL-6 & IL-10) was more significant that than Dexamethasones affects on anti-inflammatory (IL-10), substantiating speculations that Dexamethasone inhibits inflammatory responses more than it stimulates anti-inflammatory processes.

References

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